

Day 6 — Indian Astronomy + The Yavana Exchange

Module: What is IKS? (Senior) · **Time:** 45 min

Learning objective

Students articulate three specific Indian astronomical achievements AND the Greek-Indian intellectual exchange that complicates the “purely Indian” framing.

Materials

- Printed handout: one verse from Aryabhaṭīya in translation
- Printed extract from Varāhamihira’s *Pañca-siddhāntikā* (in translation)

Flow

Warm-up (5 min)

Recall: who was Aryabhata? When did he live?

Core (25 min)

1. **Aryabhata (~476–550 CE). *Aryabhaṭīya* (499 CE).**
 - Earth rotates on its axis.
 - Approximated π to 4 decimals.
 - Computed planetary periods.
 - Did NOT propose heliocentrism.
2. **Sūrya-siddhānta (early CE). Comprehensive astronomical text.**
 - Earth’s circumference computed within ~1% of modern value.
 - Detailed eclipse predictions.
3. **Varāhamihira (~6th c. CE). *Pañca-siddhāntikā*.**
 - **Compares 5 different astronomical systems** — including a “Romaka” (Greek-derived) system and a “Paulīśa” system possibly derived from Greek astronomer Paul of Alexandria.
 - **This is huge.** It documents that 6th-c. Indian astronomers were actively reading and engaging Greek astronomical work.
4. **The Yavana question.**
 - *Yavana* = “Greek” in Sanskrit. The Indo-Greek kingdoms (2nd c. BCE – 1st c. CE) created sustained cultural-intellectual exchange.
 - Greek astronomical methods influenced Indian astronomy.
 - Indian astronomical methods (Hindu numerals, trigonometric tables) influenced Arab and via Arab, European astronomy.
 - The “pure Indian” framing of IKS doesn’t survive contact with this evidence. The honest framing: Indian intellectual tradition is *partly indigenous, partly cosmopolitan*, in dynamic exchange.

Discussion (10 min)

- “Does the Yavana exchange weaken the case for celebrating Indian astronomical achievements?”
- The honest answer: no. *Synthesis* is also an achievement. Indian astronomers absorbed Greek methods AND developed their own (e.g. sine instead of chord, Hindu numerals).

Wrap-up (5 min)

- Preview Day 7: “Tomorrow — the contemporary policy debate. NEP §4.6.”

Differentiation

- **Tier up:** Read Pingree, D. (1996). “Astronomy in India” in *Astronomy Before the Telescope*.
- **Tier down:** Focus on the three achievements; the Yavana question is the depth layer.

Teacher notes

- The Yavana evidence is well-documented. Don’t shy from it. Honest history shows the Indian tradition as participant in a wider scholarly world.

Citation(s) used

- Aryabhata, *Aryabhaṭīya* (499 CE) — Clark (1930).
- Varāhamihira, *Pañca-siddhāntikā* — Neugebauer & Pingree (1971).
- Pingree (1996).